

Module 14: PostgreSQL Security

Overview

This module covers the full security stack for PostgreSQL deployments: authentication, authorization, encryption, auditing, secrets management, and compliance. Each file provides production-grade configurations, real-world patterns, and compliance guidance.

Prerequisites

- PostgreSQL administration basics (Modules 1-5)
- Basic Linux/networking knowledge
- Understanding of SQL and roles

Module Contents

File	Topic	Key Concepts
01_authentication_methods.md	pg_hba.conf, auth methods	trust/md5/SCRAM/cert/LDAP
02_roles_and_privileges.md	Roles, GRANT, REVOKE	RBAC, group roles, default privileges
03_row_level_security.md	RLS policies	7 policy patterns, tenant isolation
04_ssl_tls.md	SSL/TLS setup	sslmode, certs, mTLS, rotation
05_encryption.md	pgcrypto, column encryption	bcrypt, PGP, symmetric, TDE
06_auditing.md	pgaudit, change history	DDL audit, trigger-based audit
07_secrets_management.md	Vault, .pgpass, rotation	Dynamic credentials, cert lifecycle
08_compliance_considerations.md	GDPR, SOC2, HIPAA	Erasure, PHI, evidence collection

Security Layers at a Glance

Network:	SSL/TLS (verify-full) — Firewall — VPN
Auth:	scram-sha-256 — LDAP — Certificate
AuthZ:	RBAC roles — Column privileges — RLS
Audit:	pgaudit — Triggers — Event triggers
Encrypt:	pgcrypto — Filesystem — Backup
Secrets:	Vault — SecretsManager — .pgpass

Quick Start: Minimum Security Configuration

```
-- 1. Use scram-sha-256 (pg_hba.conf: hostssl ... scram-sha-256)
-- 2. Revoke public schema creation
REVOKE CREATE ON SCHEMA public FROM PUBLIC;
```

```
-- 3. Create application roles (not superuser)
CREATE ROLE appuser WITH LOGIN PASSWORD 'Strong!' CONNECTION LIMIT 50;
-- 4. Enable RLS on sensitive tables
ALTER TABLE users ENABLE ROW LEVEL SECURITY;
-- 5. Install pgaudit and audit DDL
```

Related Modules

- Module 13: Replication & HA (security for replication connections)
- Module 15: Backup & Recovery (backup encryption)